

Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br
	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
	Kr	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	Cs	Ba	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
P	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103		
	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn	Fr	Ra	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr		
GA	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
	He	Ne	Ar	Kr	Xe	Rn	Fr	Cs	Rb	K	Na	Li	Ra	Ba	Sr	Ca	Yb	Eu	Y	Sc	Lu	Tm	Er	Ho	Dy	Tb	Gd	Sm	Pm	Nd	Pr	Ce	La	Lr	No
	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
	Md	Fm	Es	Cf	Bk	Cm	Am	Pu	Np	U	Pa	Th	Ac	Zr	Hf	Ti	Nb	Ta	V	Mo	W	Cr	Tc	Re	Mn	Fe	Os	Ru	Co	Ir	Rh	Ni	Pt	Pd	Au
Pm	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103		
	Ag	Cu	Mg	Hg	Cd	Zn	Be	Tl	In	Al	Ga	Pb	Sn	Ge	Si	B	Bi	Sb	As	P	Po	Te	Se	S	C	At	I	Br	Cl	N	O	F	H		
Pm	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
	He	Ne	Ar	Kr	Xe	Rn	Fr	Cs	Rb	K	Na	Li	Ra	Ba	Sr	Ca	Eu	Yb	Lu	Tm	Y	Er	Ho	Dy	Tb	Gd	Sm	Pm	Nd	Pr	Ce	La	Ac	Th	Pa
	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	Sc	Zr	Hf	Ti	Ta	Nb	V	Cr	Mo	W	Re	Tc	Os	Ru	Ir	Rh	Pt	Pd	Au	Ag	Cu	Ni	Co
	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103		
	Fe	Mn	Mg	Zn	Cd	Hg	Be	Al	Ga	In	Tl	Pb	Sn	Ge	Si	B	C	N	P	As	Sb	Bi	Po	Te	Se	S	O	At	I	Br	Cl	F	H		

alkali metal

alkaline earth metal

metal

halogen

metalloid

noble gas

nonmetal

transition metal

lanthanides

actinides